

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR18

Fibrinolysis and Disorders of Hemostasis

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Fibrinolysis and disorders of hemostasis

[Coagulation Cascade Animation - Physiology of Hemostasis - YouTube](#);
[Platelet Activation and Factors for Clot Formation - YouTube](#)

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Lippincott's Biochemistry 8th Edition – Chapter 35 (online access only, online Q for the lecture)

[**Medscape: Coagulation Testing in the Core Laboratory**](#)

Practice questions for Hemostasis on Sakai

Concept map: Lippincott's Figure 35.25

Lecture: Fibrinolysis and Disorders of Hemostasis

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0251	Indicate the role of calcium in hemostasis
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0256	Discuss the role of vitamin K in the post-translational modification of clotting factors
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0257	Predict the clinical manifestations in a patient with vitamin K deficiency
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0258	Outline the anticoagulant and fibrinolytic roles of PGI ₂ , antithrombin III, proteins C and S, and plasmin.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0259	Appraise the role of bleeding time, platelet function tests (ristocetin cofactor assay), prothrombin time (INR), and activated partial thromboplastin time (APTT) in the differentiation of bleeding disorders.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0260	Indicate the mechanism of action of aspirin, warfarin, Tissue plasminogen activator and heparin
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0261	Discuss the molecular basis and laboratory findings in the following disorders of hemostasis: Hemophilia (A and B), thrombocytopenia, von Willebrand Disease, Bernard-Soulier Syndrome, Glanzmann thrombasthenia.

Recommended Reading

- Lippincott's Biochemistry 9th Edition – Chapter 35 (Practice Questions) [Blood Clotting | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Ref: Robbins Basic Pathology (11th Ed), Chapter 3 [Hemodynamic Disorders, Thromboembolism, and Shock - Robbins & Kumar Basic Pathology - ClinicalKey Student](#) – Section: Normal hemostasis
- Practice questions for Hemostasis on Sakai
- Concept map: Lippincott's Figure 35.25

Key Facts of Hemostasis

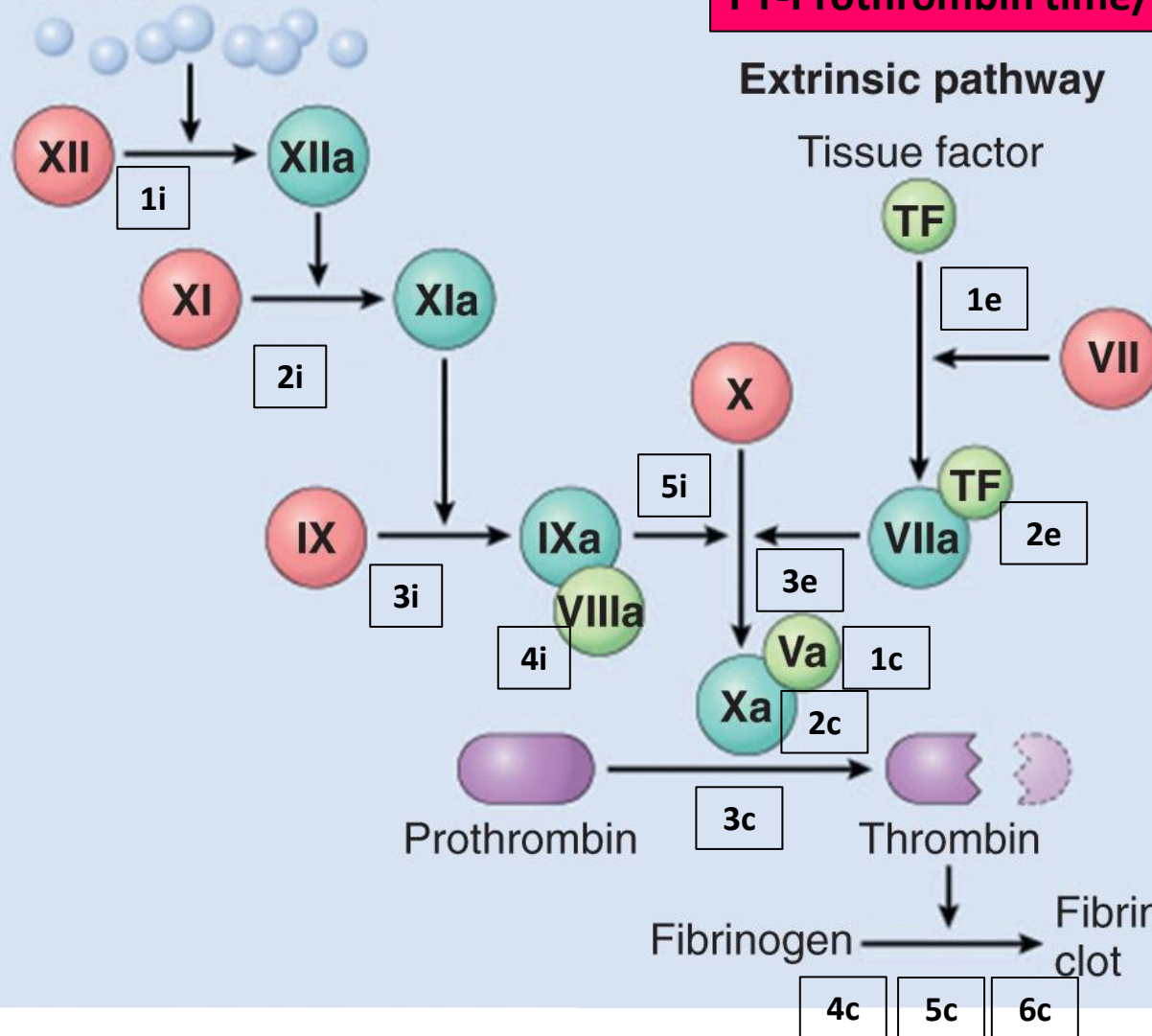
- Platelet plug formation requires: GPIb, GPIIb-IIIa, GPIa, vWF, platelets
 - **Laboratory test for platelet plug formation:**
 - **Bleeding time; Platelet aggregation tests**
- Coagulation cascade:
 - Factor Xa generated by
 - **Extrinsic pathway (1e-3e) – through VII and III (*activated on injury*)**
 - **Intrinsic pathway (1i-5i) – through XII, XI, IX and VIII**
 - Factor Xa generates thrombin
 - Thrombin converts fibrinogen to fibrin (4c) and stabilizes fibrin (6c)
 - **Laboratory tests for coagulation cascade:**
 - **Prothrombin time (INR): Extrinsic pathway**
 - **Activated Partial Thromboplastin time (aPTT): Intrinsic pathway**

APTT

CLOTTING IN THE LABORATORY

Intrinsic pathway

Negatively charged surface
(e.g., glass beads)



PT-Prothrombin time/ INR

Extrinsic pathway

Tissue factor

Factor I – Fibrinogen
Factor II- Prothrombin
Factor III-Tissue factor
Factor IV- Calcium
Factor V
Factor VII
Factor VIII
Factor IX
Factor X
Factor XI
Factor XII
Factor XIII

LEGEND:

Green – Common pathway factors

Blue – Extrinsic pathway factors

Black – Intrinsic pathway factors

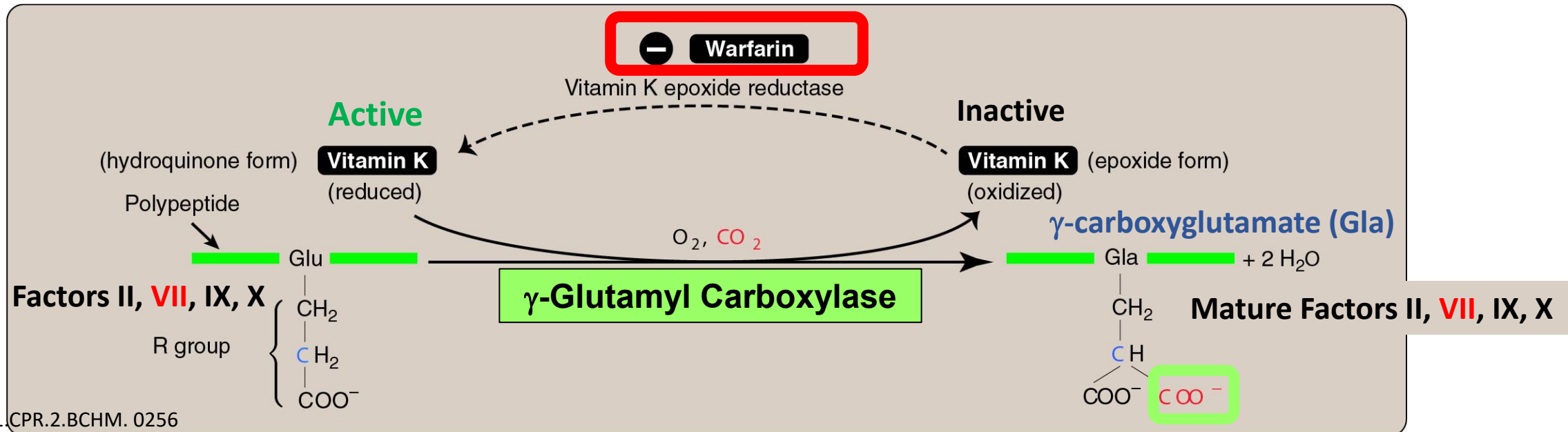
Clotting factor song:

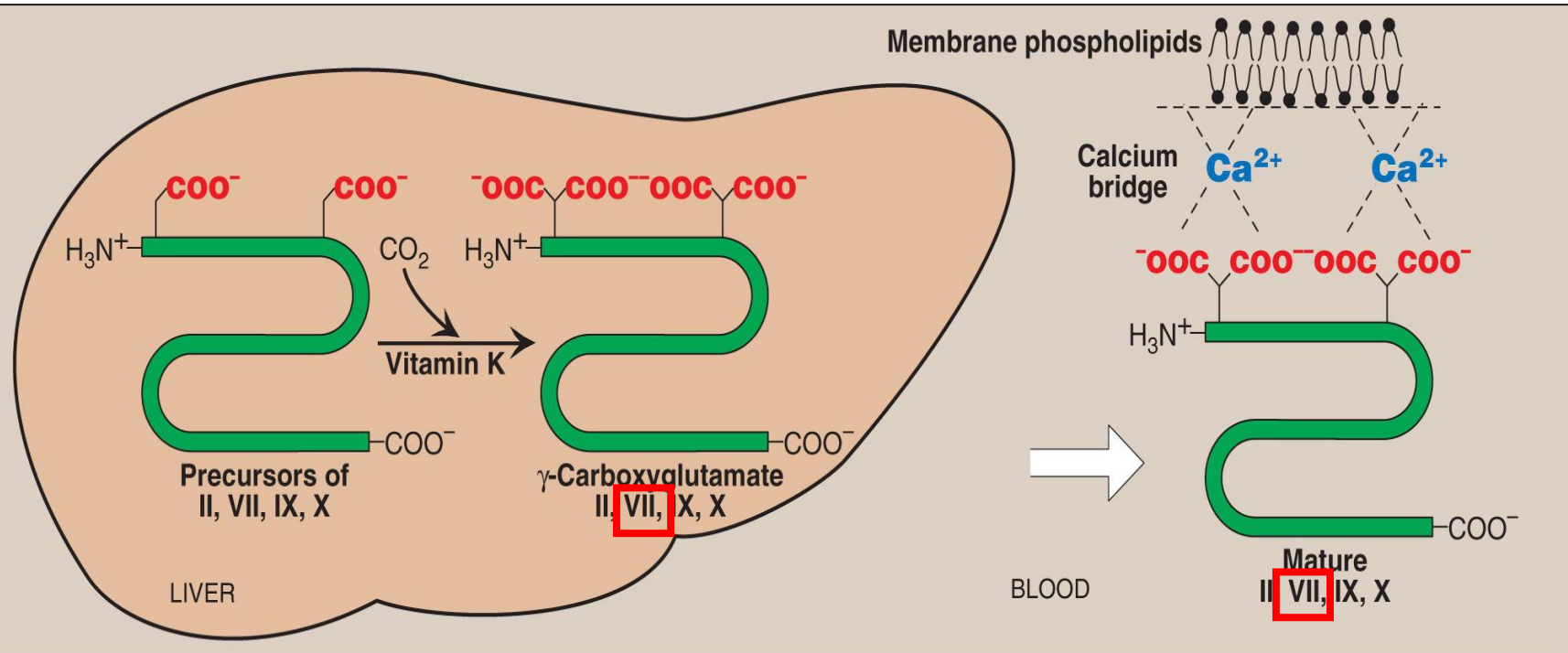
<http://www.youtube.com/watch?v=TG9yPHH3FmY>

A

Role of vitamin K and γ -Carboxylation

- Vit K dependent: **Prothrombin (Factor II)**, **VII**, IX, X, Proteins C and S
- **γ -carboxylation of glutamic acid** residues of these proteins
- Vit K (fat soluble vitamin) **coenzyme** for **γ -Glutamyl carboxylase**
- Post-translational modification forms mature clotting factors that contain **γ -carboxyglutamate (Gla)** → Released by liver and participate in clotting
- **Warfarin** inhibits **VKOR (vitamin K epoxide reductase)**
- **VKOR**: Regenerates vitamin K



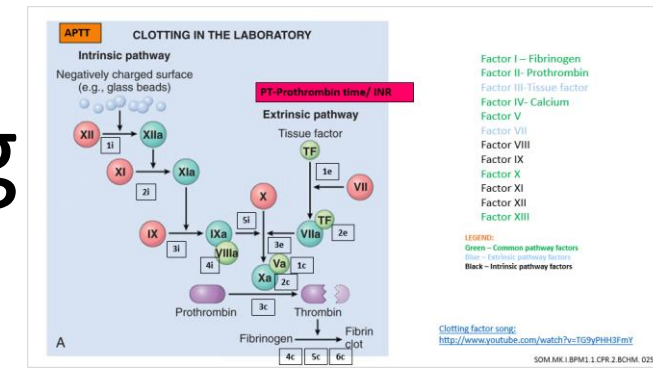


Role of vitamin K & γ -Carboxylation

- Allows **Ca^{2+} binding** because of two negatively charged carboxylate groups
- Clotting factor- Ca^{2+} complex binds to phospholipids on platelet membranes
- **INR (Prothrombin time/PT)**: Very sensitive indicator of vitamin K deficiency and for follow-up of patients on warfarin therapy. **Prolonged Prothrombin time (INR)**
 - **Factor VII (extrinsic pathway)** levels most sensitive to vitamin K

Role of calcium in clotting

- Calcium ions (Factor IV)
- Calcium ions required for all steps that require vitamin K dependent clotting **factors II, VII, IX and X**
- **Calcium chelators (EDTA/ oxalate)** added to blood in vitro to prevent clotting and maintain blood in fluid state (Anticoagulant) → After centrifugation, supernatant is **PLASMA**
- **PLASMA: has all coagulation factors**
- **SERUM: supernatant obtained after clotting (contains no coagulation factors)**



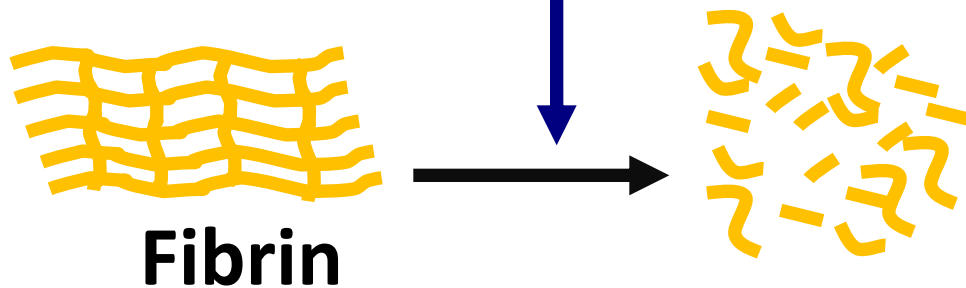
4. Fibrinolysis: Dissolution of fibrin clot (Tertiary hemostasis)

Inactive Plasminogen incorporated in clot

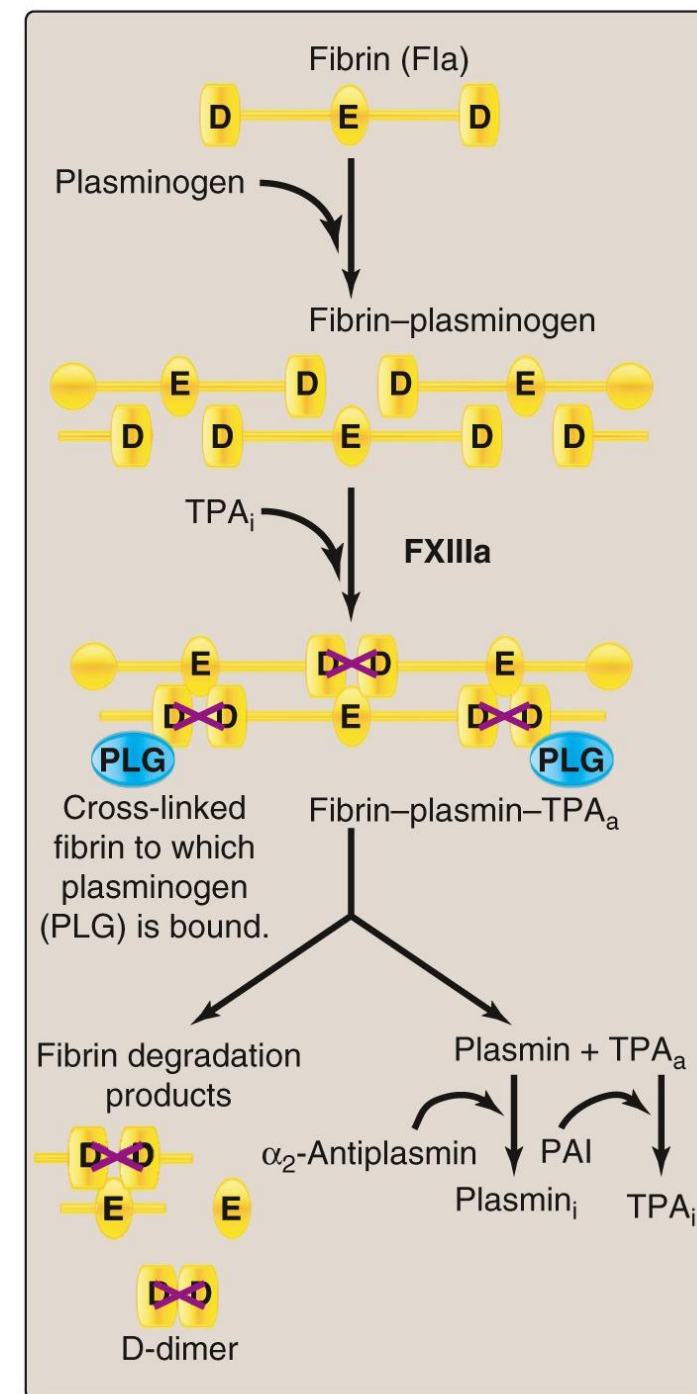
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1. Tissue Plasminogen Activator
2. Streptokinase/ Urokinase

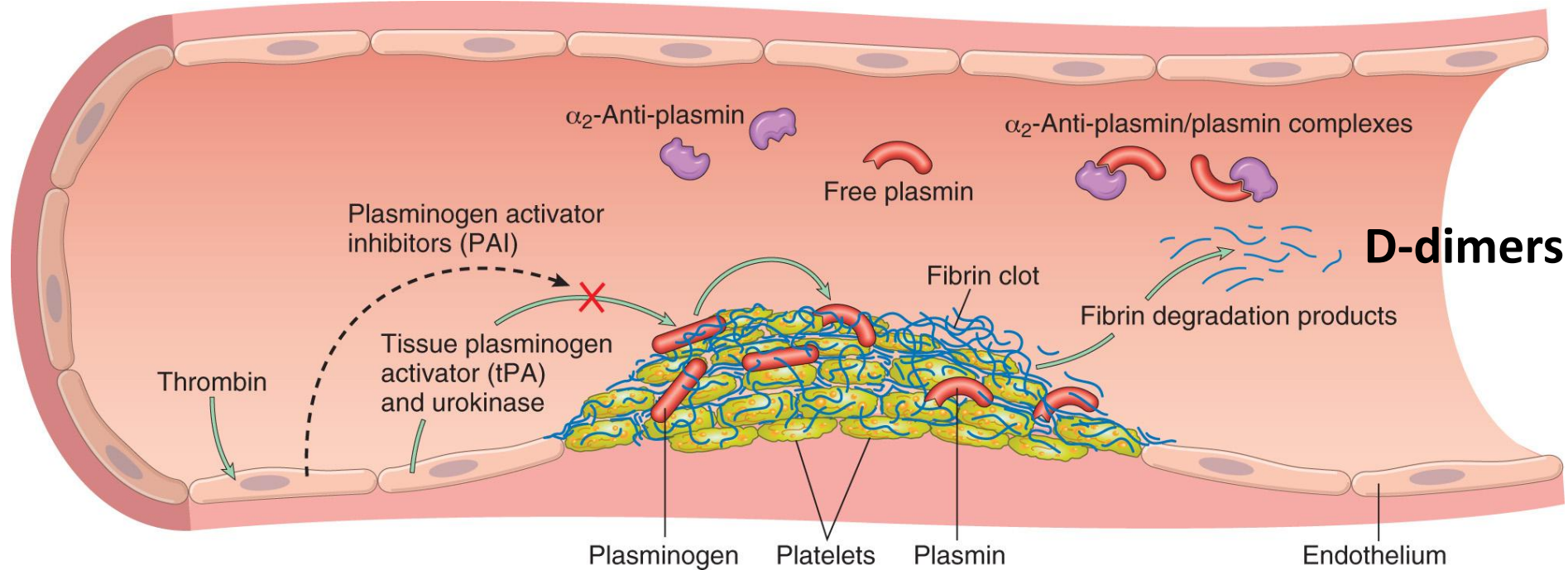
Active Plasmin (proteolytic)



Fibrin degradation
Products (FDP)
D-dimers



4. Fibrinolysis: Dissolution of fibrin clot (tertiary hemostasis)



D-dimer levels elevated in **deep vein thrombosis**.

Estimate extent and rate of fibrinolysis

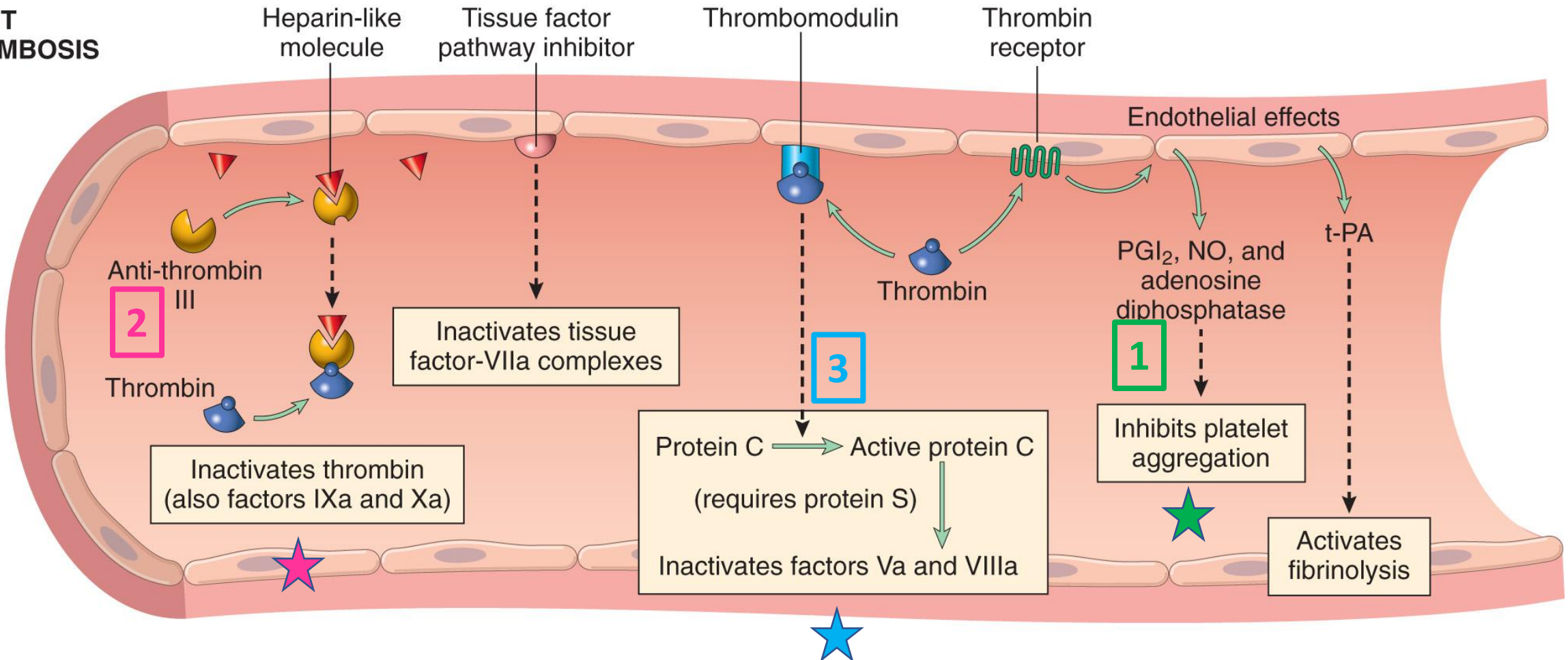
Highly elevated D-dimer levels indicate high risk of **pulmonary embolism**

Control of hemostasis

- Endothelial cells are central regulators of hemostasis
- Continuous blood flow prevents clotting
- Normal endothelia are anti-thrombotic
- Coagulation automatically initiates **fibrinolysis**
- **Balance between anti-thrombotic and prothrombotic factors**

Anti-coagulant factors

INHIBIT THROMBOSIS



Anti-coagulant factors

1. Endothelial PGI₂ (prostacyclin) and nitric oxide prevent platelet aggregation

- PGI₂ - increases platelet cAMP levels and inhibits activation (Thromboxane antagonist)

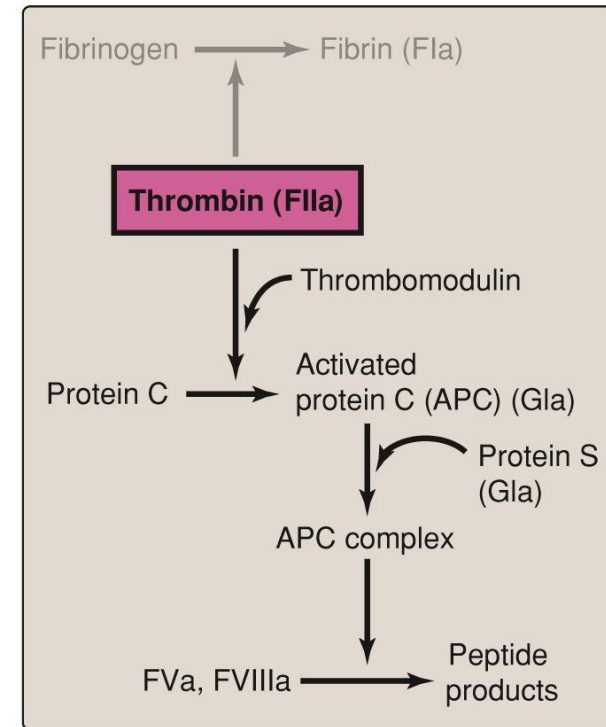
2. Antithrombin-III

- Inactivates **thrombin** and **factor Xa** and prevents clotting
- **Heparin (glycosaminoglycan)** activates Antithrombin-III

3. Protein C and S (Vit K dependent)

- Thrombin + **Thrombomodulin** → Activates Protein C and protein S
- Proteins C and S → Inactivate **Va and VIIIa** of coagulation cascade

What happens if anticoagulant system is deficient??



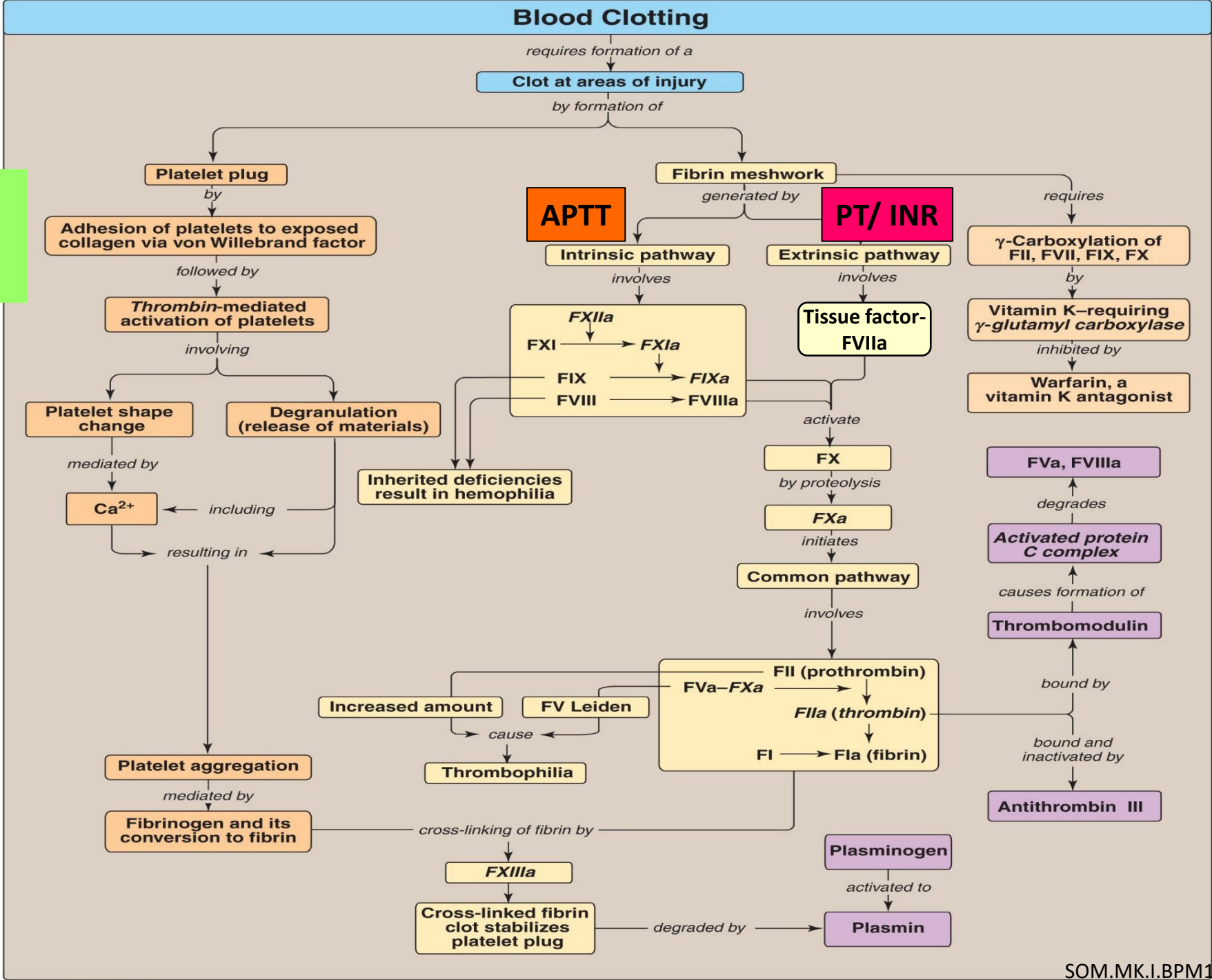
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A bit of pharmacology.....

- **Aspirin:** Inhibits platelet thromboxane (TXA₂) formation
 - Irreversible inhibitor of **cyclooxygenase (COX)**
 - Decreases TXA₂: PGI₂ ratio
- **Heparin:** Activates **antithrombin III** and inactivates thrombin
- **Warfarin (oral anticoagulant):** Blocks **vitamin K epoxide reductase (VKOR)** in liver and prevents regeneration of active form of Vitamin K
 - Inhibits synthesis of mature vitamin-K dependent clotting factors (II, VII, IX and X)
- **Streptokinase/ Urokinase:** Thrombolytic agent; **plasminogen activator**;
 - Converts plasminogen to plasmin enabling dissolution of clots
- **Tissue Plasminogen activator:** Used to dissolve a thrombus
 - Converts plasminogen to plasmin enabling dissolution of clots

Bleeding time
Platelet aggregation
Platelet count

Lippincott’s Figure 35.25



Tests for platelet function/aggregation

Bleeding time (BT)

- Time taken from **initial injury to formation of platelet plug**
- Indicator of **platelet plug formation (primary hemostasis)**
- **Not a reliable test**
- Prolonged bleeding time is an indicator of
 - Low platelet count OR
 - vWf deficiency OR
 - Platelet receptor defects

- **Platelet count**
- **Platelet aggregation/ function tests** measure platelet aggregation in presence of activators (ADP, thrombin, and ristocetin)
- **Ristocetin cofactor assay:** von Willebrand disease. Reduced platelet aggregation in von Willebrand disease
- **Flow cytometry:** Platelet glycoprotein defects

Tests for coagulation disorders

Prothrombin time (PT) or International normalized ratio (INR)

- Tissue factor, phospholipids, and calcium added to blood and time for formation of fibrin clot is measured
- Tests **extrinsic** and common pathways
- Measures defects in
 - **Tissue factor**
 - **Factor VII**
 - Factor V
 - Factor X
 - Prothrombin
 - Fibrinogen

Activated Partial thromboplastin time (APTT/ aPTT/ PTT)

- Activate factor XII (glass beads) along with phospholipids and calcium, and time to form fibrin clot is measured
- Tests **intrinsic** and common pathways
- Measures defects in
 - **Factor XI**
 - **Factor XII**
 - **Factor VIII**
 - **Factor IX**
 - Factor V
 - Factor X
 - Prothrombin
 - Fibrinogen

Disorders associated with defects in hemostasis

- Bleeding disorders (defects in primary or secondary hemostasis) present with **increased bruising**, increased **bleeding following surgery, epistaxis** (nose bleeds), hematomas
- Disorders of **clotting pathway (coagulation cascade)**
 - Hemophilia A and B
 - Vitamin K deficiency (review in vit K)
- Defects in **platelet plug formation**
 - Von Willebrand disease
 - Platelet defects
 - Thrombocytopenia
 - Bernard-Soulier syndrome
 - Glanzmann thrombasthenia

A 16-year-old boy comes to the physician with a history of bleeding gums and bleeding from his nose. He had similar episodes in the past. Examination shows purpuric spots on his skin (small bleeds in the skin)
Lab investigations:

Tests for primary hemostasis/platelet plug formation:

Bleeding time

Platelet count

vWF levels

Platelet aggregation studies - ADP, thrombin, ristocetin

Flow cytometry

Tests for secondary hemostasis/clotting cascade:

Prothrombin time (PT/INR)

APTT



Hemophilia A and B

- Inherited coagulation disorder – defect in **intrinsic coagulation pathway**
- **X-linked recessive** inheritance (affects males and females are carriers)
- Hemophilia A and B: Similar presentation and difficult to distinguish clinically
 - Individual factor assays to differentiate between them
- Clinical features (common to both)
 - Easy bruising
 - Massive hemorrhage after trauma and surgical procedures
 - Spontaneous hemorrhages, particularly in joints - hemarthrosis

<https://www.hemophilia.org/John-Gale>

**Coagulation
factor deficiency**

Hemophilia A
Factor VIII

Hemophilia B
Factor IX



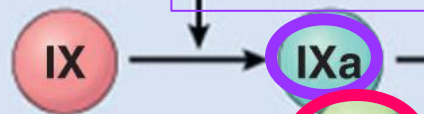
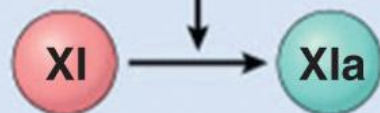
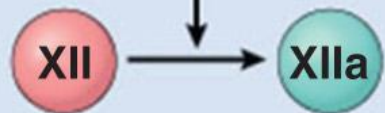
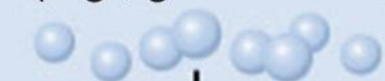


APTT - Prolonged

NG IN THE LABORATORY

Intrinsic pathway

Negatively charged surface
(e.g., glass beads)



Hemophilia A

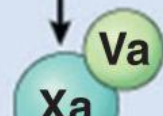
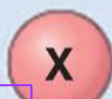
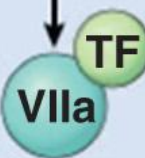
Hemophilia B



Prothrombin

Extrinsic pathway

Tissue factor



Thrombin

Fibrinogen

Fibrin
clot

A

PT-Prothrombin time/ INR - Normal



*Hemarthrosis (bleeding
into joint cavity)*



Lab findings in hemophilia A and B

- **Tests for primary hemostasis:**

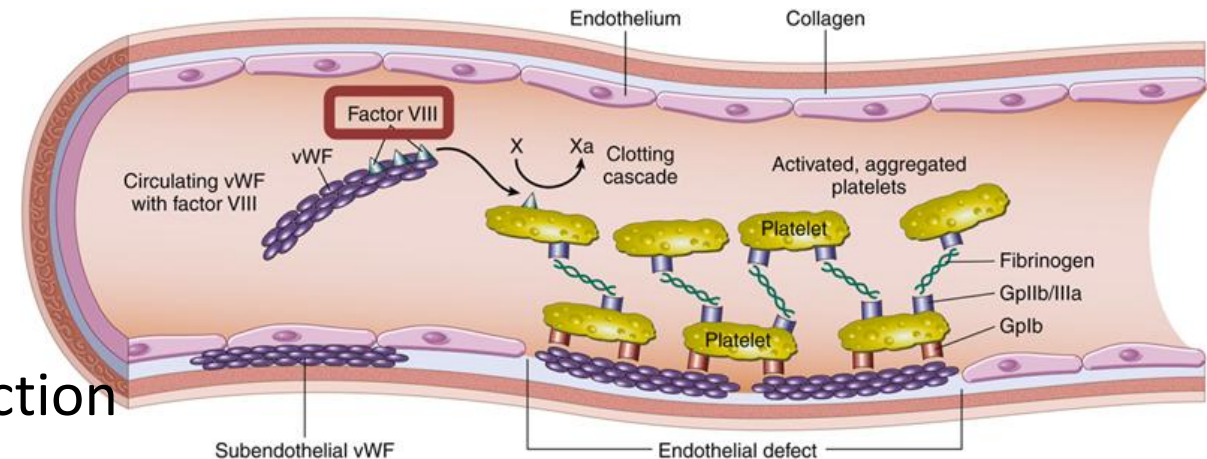
- Bleeding time: **Normal** (Platelet plug formation is normal)
- Platelet count: **Normal**

- **Tests for secondary hemostasis:**

- **Prothrombin time (INR): Normal** (extrinsic and common coagulation pathway are normal)
- **APTT : *Prolonged* (Defect in intrinsic pathway)**
- **Individual factor assays:** To determine type of hemophilia
 - Factor VIII levels are low in hemophilia A
 - Factor IX levels are low in hemophilia B

Von Willebrand Disease

- Most common inherited bleeding disorder
- Deficiency of vWF
- Defect in **platelet plug formation (Prolonged bleeding time)**
- Also have **instability of Factor VIII** (may result in increased APTT)
 - Remember: vWF binds to and stabilizes factor VIII in circulation
- Clinical features: Clinical presentation similar to hemophilia
 - <https://www.hemophilia.org/Shelly-Mattson>
 - Increased mucosal bleeding
 - Easy bruising
 - Epistaxis (Bleeding from nose)
 - Increased post-operative bleeding
 - Prolonged bleeding following tooth extraction



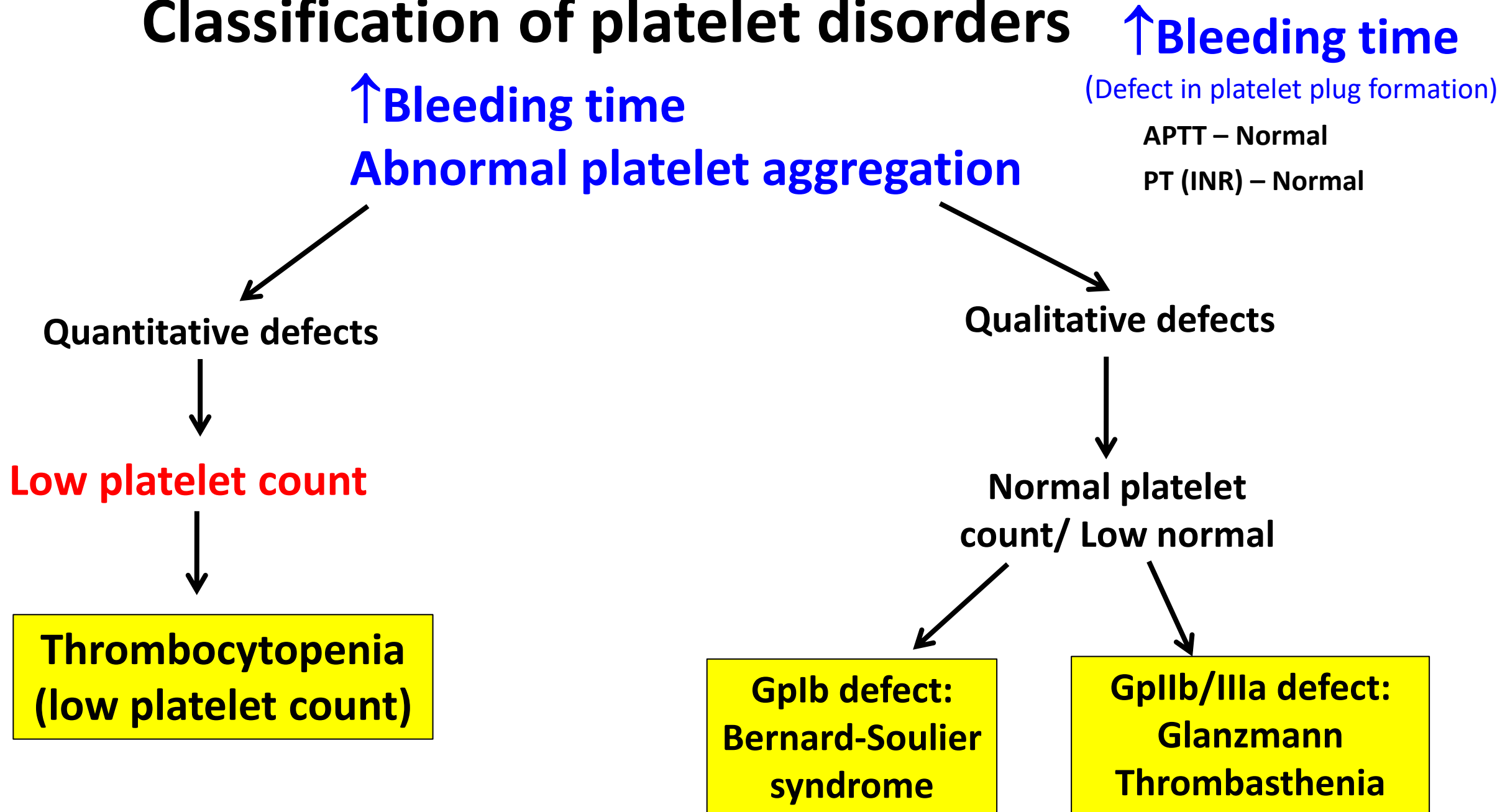
Lab Findings in von Willebrand Disease

- Tests for primary hemostasis:
 - **Bleeding time – Prolonged** (Defect in platelet plug formation)
 - Platelet count – Normal
 - **Ristocetin assay** for platelet aggregation: Abnormal
 - vWF levels – LOW
- Tests for secondary hemostasis:
 - **APTT – Prolonged** (Factor VIII levels may be low normal)
 - PT (INR) – Normal (Extrinsic pathway is normal)

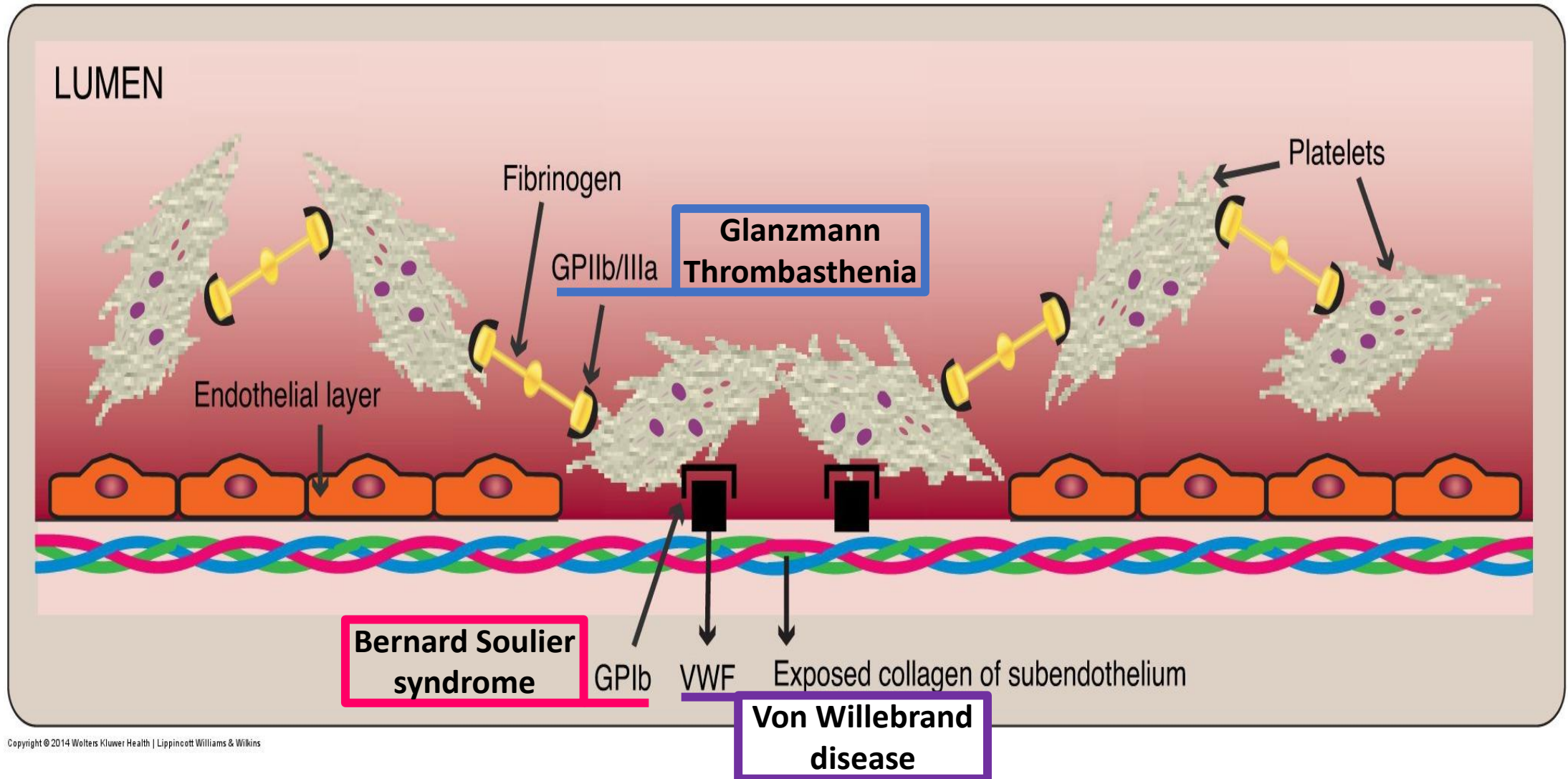
Disorders of hemostasis due to **Platelet defects**

- **Prolonged bleeding time** (Defect in formation of platelet plug)
- **Platelet aggregation tests are abnormal**
- Platelet count is obtained
- Thrombocytopenia: Increased bleeding time and **low platelet count**
- If platelet count is low normal (or within normal limits), flow cytometry to identify **platelet receptor defects**
 - GpIb defect (Bernard-Soulier syndrome)
 - GpIIb/IIIa defect (Glanzmann Thrombasthenia)
- **Tests for secondary hemostasis:**
 - APTT – Normal
 - PT (INR) – Normal

Classification of platelet disorders



Inherited platelet defects



• Figure 35.24 (Lippincott's)

A 16-year-old boy comes to the physician with a history of bleeding gums and bleeding from his nose. He has had similar episodes in the past. Examination shows purpuric spots on his skin (small bleeds in the skin)
Lab investigations were as follows:

Bleeding time - Prolonged (defect in platelet plug formation)

Prothrombin time - normal (Extrinsic and common pathways are normal)

APTT - normal (Intrinsic and common pathways are normal)

vWF levels - normal (Required for platelet plug formation)

Is this a platelet disorder or coagulation disorder?

Platelet disorder, as coagulation tests are normal - INR (PT) and APTT are normal

What other investigations can you carry out?

Platelet count - If low (thrombocytopenia)

Platelet aggregation tests

Flow cytometry to look for surface glycoprotein defects

	Disorder	Biochemical defect	Abnormal lab tests		
			BT	APTT	PT (INR)
Clotting pathway defects	Vitamin K deficiency				
	Hemophilia A				
	Hemophilia B				
	Von Willebrand disease				
Platelet plug defects	Bernard Soulier syndrome				
	Glanzmann Thrombasthenia				
	Thrombocytopenia				

Proteins/ Enzymes required for platelet plug formation	Biological role	Disorder associated
GPIb	Binds activated platelet to subendothelial collagen via vWF	Bernard Soulier syndrome: Prolonged Bleeding time; Abnormal platelet aggregation
Von Willebrand factor (vWF)	- Binds activated platelet to subendothelial collagen - Stabilizes Factor VIII	von Willebrand disease: Prolonged bleeding time; Abnormal platelet aggregation and abnormal ristocetin cofactor assay and Prolonged APTT
GPIIb/IIIa	Platelet aggregation via fibrinogen	Glanzmann thrombasthenia: Prolonged bleeding time; Abnormal platelet aggregation
Platelets	Platelet plug formation	Thrombocytopenia; Prolonged bleeding time; Abnormal platelet aggregation

Proteins/ factors Secondary hemostasis	Biological role	Disorder associated
Tissue factor	Activates extrinsic pathway	
Factor VII	- Activated by Tissue factor - VII + Tissue factor activates factor X (extrinsic pathway) + calcium and phospholipids	Vitamin K deficiency Prolonged PT
Factors XII, XI	Activates intrinsic pathway	
Factor VIII	- Factors VIII and IX activate factor X (Intrinsic pathway) + calcium and phospholipids - Von Willebrand Factor stabilizes factor VIII - Inactivated by Proteins C and S	Hemophilia A: Prolonged APTT Also, in vWF deficiency (due to unstable factor VIII)
Factor IX	- Factors VIII and IX activate factor X (Intrinsic pathway) + calcium and phospholipids	Hemophilia B: Prolonged APTT
Factor X	- Convergence of Intrinsic and extrinsic coagulation cascade - Activates prothrombin to thrombin	
Factor V	- Factor X, V, calcium and phospholipids (Prothrombinase complex) - Inactivated by proteins C and S	
Thrombin (Factor II)	- Converts fibrinogen to fibrin - Activates Factor XIII (cross-links fibrin) - Activates factors V and VIII - Activates platelet plug - Activates thrombomodulin → Activates Protein C and S	
Fibrinogen	- Converted to fibrin by thrombin	
Factor XIII	- Converts fibrin (Soft clot) → Cross-linked fibrin (Stable clot)	
Calcium (Factor IV)	- Required for conversion of prothrombin to thrombin and factor X activation	

Anticoagulant/ Fibrinolysis	Biological function
Antithrombin-III	Anticoagulant; Inactivates thrombin and factor X
Heparin	Activates antithrombin-III AT-III in turn, Inactivates thrombin and factor X
Plasminogen	Activated to plasmin and cleaves fibrin into Fibrin degradation products (FDP; D-dimers)
Tissue plasminogen activator (TPA)/ Streptokinase/ Urokinase	Activates Plasminogen to plasmin Breaks down fibrin clot to FDP
Proteins C and S	- Activated by thrombin-thrombomodulin - Inactivate Va and VIIIa
Warfarin (Coumarin)	Inhibits gamma-carboxylation of vitamin K dependent (II, VII, IX and X) clotting factors; Inhibits regeneration of vitamin K; Inhibits vitamin K epoxide reductase
EDTA (in-vitro)	Calcium chelator